

Failure Propagation in Multi-Agent LLM Systems: Characterization and Containment with Circuit Breakers

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Abstract

Large language models (LLMs) are increasingly deployed as compositional systems in which multiple agents interact to solve complex tasks. While such architectures enable modularity and improved capability, they introduce a new class of failure modes: errors originating in one component can propagate through the system, leading to cascade failures.

In this work, we study failure propagation in multi-agent LLM pipelines and evaluate circuit breakers as a mechanism for containment. We formalize cascade failures through a probabilistic model of upstream error propagation and introduce metrics such as Cascade Failure Rate (CFR) to quantify system behavior.

We conduct a large-scale simulation study (100,000+ trials) across varying pipeline depths and failure conditions, demonstrating that circuit breaker mechanisms reduce cascade failures by 50–89%. We complement these results with real-world API-based experiments using a planner–executor–verifier pipeline, where we observe that up to 62% of upstream failures propagate to final task failure without protection.

We further analyze adaptive circuit breaker strategies and find that naive adaptive policies can over-trigger, highlighting the challenges of designing robust control mechanisms in LLM systems. Our results suggest that failure containment is a critical systems concern for compositional AI and that circuit breaker-inspired mechanisms provide a promising direction for improving reliability.

1 Introduction

Large language models (LLMs) are increasingly used as building blocks in compositional systems, where multiple agents interact through structured interfaces to perform complex tasks. These systems include planner–executor pipelines, tool-augmented agents, and multi-agent reasoning frameworks. While such architectures enable modularity and specialization, they also introduce new system-level failure modes.

A key challenge in these systems is failure propagation: errors originating in upstream components can propagate through downstream stages, leading to cascade failures. Unlike isolated errors, these failures compound across stages and are harder to detect and mitigate.

We frame compositional LLM systems as reliability-critical distributed systems, where failure propagation—not just individual model error—becomes the dominant challenge. In this work, we study failure propagation as a first-class systems phenomenon and adapt circuit breaker mechanisms from distributed systems to contain such failures.

Contributions

- We formalize failure propagation in compositional LLM systems and introduce Cascade Failure Rate (CFR) as a metric for quantifying error amplification.
- We adapt circuit breaker mechanisms to LLM-based agents using inferred signals such as output validity and confidence.
- We provide both large-scale simulation and real-world API validation, showing that circuit breakers can eliminate cascade failures (CFR: $0.62 \rightarrow 0.00$).
- We identify a tradeoff between failure containment and task success, and show that naive adaptive policies can over-trigger.

2 Theoretical Framework

We model a compositional LLM system as a sequence of agents, where each stage transforms an input and may introduce or propagate errors.

Let p_u denote the probability of an *upstream failure*, i.e., the probability that an intermediate artifact becomes structurally invalid, semantically degraded, or otherwise unsuitable for downstream consumption.

Let p_p denote the *propagation probability*, i.e., the probability that an upstream failure is not corrected and continues to affect downstream stages.

Under this abstraction, the probability of final failure for a single-stage propagation can be expressed as:

$$P(\text{final failure}) = p_u \cdot p_p \quad (1)$$

We define the Cascade Failure Rate (CFR) as:

$$\text{CFR} = P(\text{final failure} \mid \text{upstream failure}) \quad (2)$$

CFR captures the conditional likelihood that an upstream failure propagates through the system to produce an end-to-end failure.

Observation. As pipeline depth increases, the number of opportunities for propagation grows. Under repeated composition, this leads to amplification of failure probability across stages, even when individual stage failure rates remain constant.

Proposition. Assuming independent propagation across stages, the probability that a failure propagates through a pipeline of length L is:

$$\text{CFR}(L) = 1 - (1 - p_p)^L \quad (3)$$

which is monotonically increasing in L .

Implication. This formulation highlights a key systems insight: even moderate propagation probabilities can lead to high end-to-end failure rates in deeper pipelines. As L increases, the system becomes increasingly sensitive to upstream errors, making failure containment essential.

Circuit breakers can be interpreted as mechanisms that reduce the effective propagation probability p_p by intercepting faulty intermediate states. In this framework, containment strategies operate by transforming the propagation process itself rather than reducing upstream error rates p_u .

3 Method

We evaluate three configurations:

- **No Circuit Breaker**
- **AI-Based Circuit Breaker**
- **Adaptive Circuit Breaker**

The AI-based circuit breaker uses structural validity, missing fields, and confidence signals. The adaptive variant dynamically adjusts intervention thresholds.

3.1 Simulation Setup

We simulate pipelines of depth 2–6 with probabilistic failure injection. Each agent introduces failures with probability p_u , and propagation occurs with probability p_p .

3.2 Real-World Setup

We implement a planner–executor–verifier pipeline using a commercial LLM API.

An *upstream failure* is defined as an intermediate output that is structurally invalid, incomplete, or low-confidence. A *cascade* occurs when such a failure leads to final task failure without being intercepted.

4 Simulation Results

Without circuit breakers, CFR increases sharply with depth. AI-based breakers suppress propagation, while adaptive breakers achieve the strongest containment.

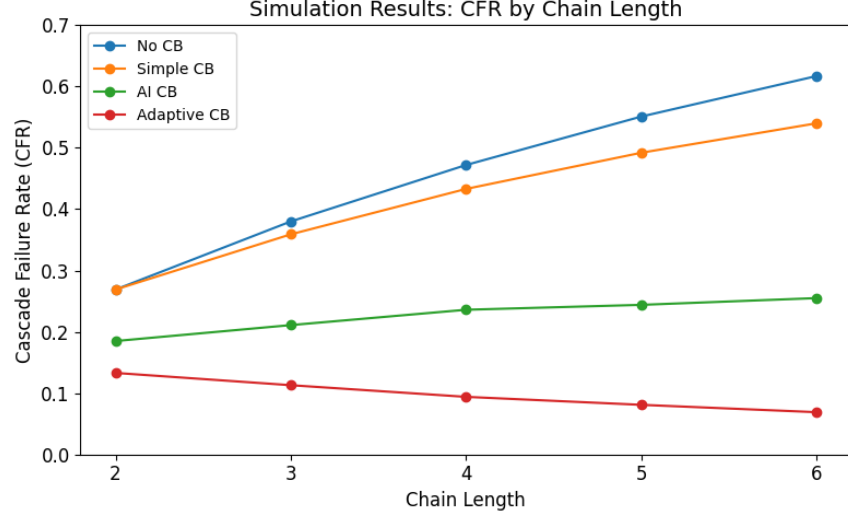


Figure 1: CFR vs chain length

5 Real-World API Validation

Without protection, 62% of upstream failures propagate. Circuit breakers eliminate cascades and improve success rates under moderate perturbation.

Method	Cascade Rate	Success
No CB	0.62	0.38
AI CB	0.00	0.44
Adaptive CB	0.00	0.40

Table 1: Real-world results

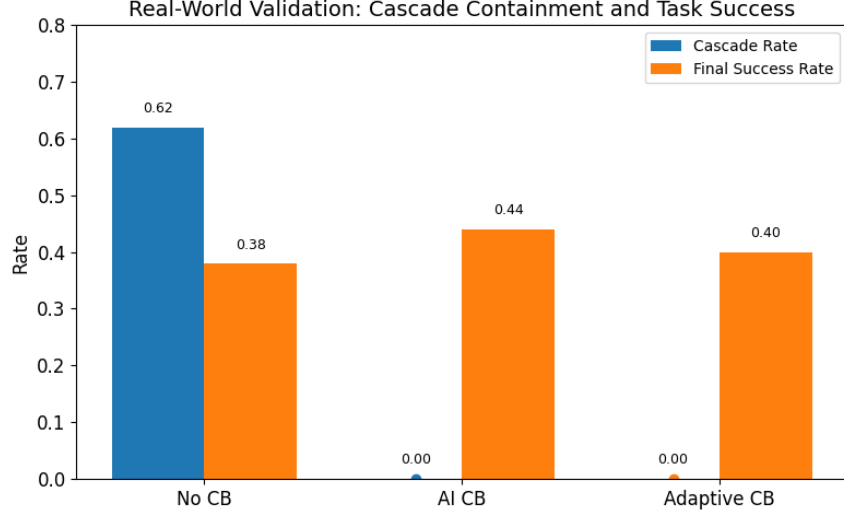


Figure 2: Real-world validation

6 Tradeoff Analysis

Adaptive policies eliminate cascades but over-trigger, reducing success. Static AI-based breakers provide better balance.

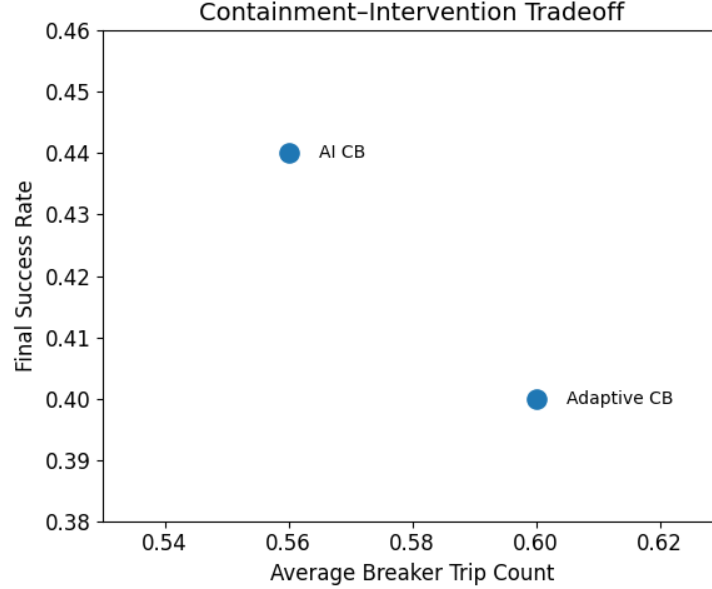


Figure 3: Containment vs intervention tradeoff

7 Discussion

Failure propagation scales with pipeline depth. Circuit breakers are effective but introduce tradeoffs between containment and task success.

8 Limitations

We evaluate relatively simple pipelines and synthetic failures. Adaptive strategies are heuristic.

9 Future Work

Future work should develop principled adaptive policies and extend to more complex multi-agent systems.

References